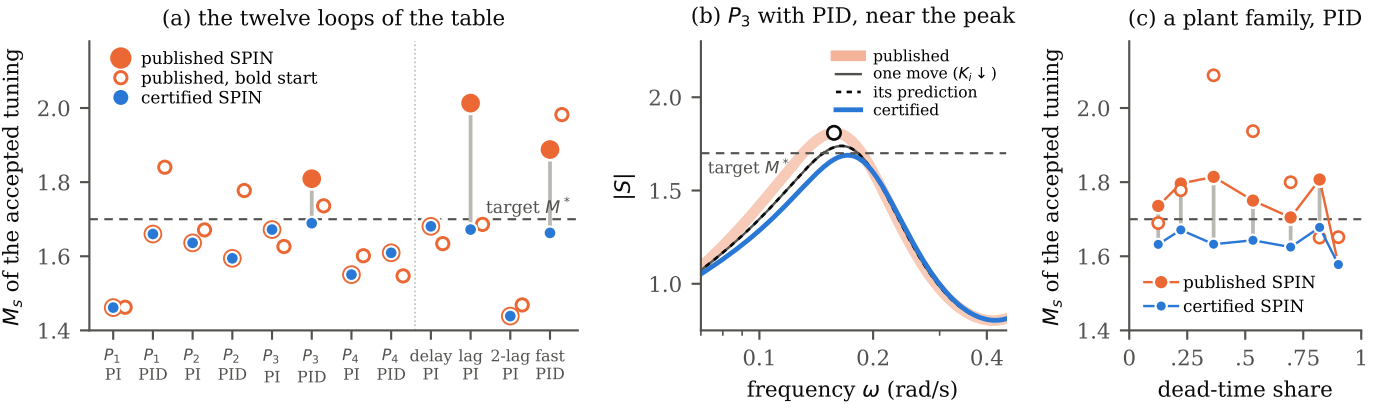
(a) the twelve loops of the table
published SPIN
published, bold start
certified SPIN
M_s of the accepted tuning
2.0
1.8
1.6
1.4
target M^*
P_1 PI
P_1 PID
P_2 PI
P_2 PID
P_3 PI
P_3 PID
P_4 PI
P_4 PID
delay lag PI
2-lag PI
fast PID
(b) P_3 with PID, near the peak
published
one move ($K_i \downarrow$)
its prediction
certified
$|S|$
2.0
1.5
1.0
target M^*
0.1
0.2
0.4
frequency ω (rad/s)
(c) a plant family, PID
M_s of the accepted tuning
2.0
1.8
1.6
1.4
published SPIN
certified SPIN
0
.25
.5
.75
1
dead-time share